

possession of such an organ to any species of animal whatsoever, as long as they had been told that two very different kinds of animals (such as dogs and bees) both possessed it. Young children's thought processes were different; they would decide whether or not to extend a property to a new kind of animal depending on that animal's perceived proximity to either of the two species (*i.e.*, depending on the strength of the analogy in question), and they did not use the superordinate category *animal* to make the analogy, because that abstract category was not sufficiently familiar to them.

Everyday Concepts Versus Scientific Concepts

A narrow vision of learning that we would be happy to see fully eliminated from educational programs presumes that knowledge acquired in school is independent of everyday knowledge. This philosophy would encourage the teaching of new ideas without any reference to everyday concepts, except for the most rudimentary notions, which are impossible to separate from everyday experience. Such a limited view of learning is based on a conception of the mind according to which we would keep facts and ideas that we pick up in school in a separate mental compartment from facts and ideas that we pick up in daily life, and year after year we would increase our school-based knowledge much as we would build a brick house: each brick added would be supported by previously installed bricks, and would in turn support new bricks.

Such a vision is appealing in some ways, since it would simplify the design of school curricula. It would make teaching easier, treating each discipline as an island disconnected from other disciplines, and trying, within any given discipline, to decompose its notions in a logical fashion and to arrange them in a strict, natural order. Teaching any complex idea would thus involve teaching a set of simpler ideas in the same domain, each of which would in turn be taught through yet simpler ideas in the same domain, and so forth.

It is a long-lived myth in the world of education that there is a watertight boundary between two types of knowledge: everyday knowledge, presumed to grow on its own with no need for formal teaching, and formal knowledge, conveyed in schools and presumed to be communicable independently of everyday knowledge. In this naïve view, school is seen as a magical shortcut that allows ideas arduously developed by humanity over thousands of years to be transmitted in just a few years to a random human being.

Another belief that exerts considerable influence on school curricula, particularly in science, is that scientific knowledge is best conveyed in precise, formal terms — especially through mathematical formulas — and that precise formalisms correspond to the way in which knowledge ought to be absorbed by beginners and in which it is manipulated by experts. It is presumed that the initially large gap between a printed formalism and one's internal mental representation gradually approaches zero as one comes closer and closer to expertise.

While we're on the subject of distances approaching zero, the following rather prickly formula expresses the idea that the function $f(x)$ is continuous at the point x_0 :

$$\forall \varepsilon \exists \delta \forall x \left| x - x_0 \right| < \delta \Rightarrow \left| f(x) - f(x_0) \right| < \varepsilon$$

(The upside-down “A” and the backwards “E” are shorthand notations for the words “for all” and “there exists”, and the arrow “ $\Rightarrow$ ” can be read as “if... then...”). Should one presume that all people who are comfortable with mathematics think of continuity in exactly these terms? Do people fluent in calculus really always imagine and mentally manipulate Greek letters, and is this what continuity means for them? Are all their prior intuitive notions of continuity totally dispensed with after they have absorbed this formula? If so, then in order to educate experts, or even to convey technical notions of this sort to ordinary students, the right way would be to transmit just such formulas, since they presumably embody the most distilled and precise essence of the notions. In particular, one would want the notion of *continuity* to be seen by students as synonymous with the above-displayed formula.

However, this view of education unfortunately conflates the actual way that experts think with the use of dense formalisms designed to capture subtle notions as rigorously and unambiguously as possible. It is the result of a long-standing philosophical assumption, reinforced by the broader culture, which is that logical thinking is superior to analogical thinking. More specifically, this view comes from misguided stereotypes of analogical thinking, which maintain that dependence on analogy, although possibly useful when one is just starting out in a field, is basically childish, and that analogies should rapidly be shed, like crutches or training wheels, when one gets down to brass tacks and starts seriously thinking in the domain. A related half-baked stereotype of analogical thinking is that it is like an untamable wild horse, so unpredictable and unreliable that it must be shunned, even if it might once in a while provide a spark of true insight; thus analogies belong not to the realm of reason but to that of “intuitions”, which, being irrational, cannot and should not be taught.

The fluid way that a scientific notion is realized and “lives” in the minds of people who deeply understand it is a very different thing from a formal and rigid symbolic notation, which has been carefully devised to be as concise and rigorous as possible, and the two should not be conflated. As we have just seen, the notion of *continuous function* has a precise formal definition, and yet continuous functions form a mental category having blurry boundaries and members with different degrees of typicality, and such judgments will vary from one mathematician to another. A famous historical case illustrates this fact. Toward the end of the nineteenth century, functions continuous at every irrational point and discontinuous at every rational point were discovered, and this behavior was so unexpected that some renowned mathematicians labeled them “pathological” and strove to banish this “scourge” from mathematics, whereas other mathematicians were extremely excited by the revelation of such a rich new area for research. This episode hardly jibes with the image of mathematics as a frozen body of precise and absolutely objective knowledge. In the following chapter we’ll see that even a category as familiar as *number* has blurry boundaries and members with varying degrees of typicality, and seasoned mathematicians can hold different opinions and can argue vehemently about what is and what is not a member of the category.

The unfortunate but widespread conflation of austere formal definitions with the psychological reality of concepts in human minds has had worrisome repercussions in education. One consequence is that it tends to make the educational establishment lose sight of what ought to be its primary goal — the construction of useful, reliable categories. Another consequence is that it tends to favor teaching methods based on prickly formalisms and rigorous deductions rather than methods that use analogies to build up suitable families of categories in intuitive ways.

Our view is very different from one in which logic is seen as central. Indeed, as we stressed in Chapter 4, expertise builds up as categories are acquired and organized. Rather than depending on formal perceptions of situations, people have the ability to treat novel situations as if they were familiar, thanks to categorization. To acquire knowledge in a domain is to build up relevant categories. Analogical thinking is the key to understanding new situations and to building up new concepts, and this holds at all levels, ranging from the shakiest beginner to the most fluent expert. The difference between those two is not their style of thinking — logical for the expert and analogical for the beginner — but the repertoire of categories that they have at their disposition, and the way those categories are organized.

For example, in the case of continuity of a function at a given point x_0 , do experts really think in terms of Greek letters and flipped and rotated roman letters? Hardly. Instead, they have vivid visual imagery about the concepts concerned. The epsilons and deltas in the formula are merely helpers in translating that imagery. Thus if one wants the value of $f(x)$ to be very close to that of $f(x_0)$ — in particular, if one wants their discrepancy not to exceed the tiny number ε (this desideratum is expressed by the notation “ $|f(x) - f(x_0)| < \varepsilon$ ”) — then continuity requires that there should exist some little zone centered on the point x_0 and whose two edges are at a distance of δ from the center (this zone is expressed by the notation “ $|x - x_0| < \delta$ ”), and throughout this zone the desideratum should be the case.

Thus if one has a mathematically trained mind and is thinking about what continuity means, one envisions a small rectangular box centered on the point $(x_0, f(x_0))$ in the plane. Continuity then will mean that no matter how close one wishes $f(x)$ to be to $f(x_0)$ (this desire translates into the image of a box whose *height* is very tiny, so that all y -values in it are *vertically* close), one can always make the box so *narrow* that the desideratum will hold everywhere inside it. The phrase “no matter how close one wishes” is the visual translation of “ $\forall \varepsilon$ ” (pronounced “for all epsilon”), while the phrase “one can always make the box so narrow” is the visual translation of “ $\exists \delta$ ” (pronounced “there exists some delta”). In the end, it all comes down to the idea of zooming in on the graph of the function at arbitrarily small scales, and thus it has to do with such familiar things as magnifying glasses and microscopes and walking towards objects so that what is blurry at a distance comes into focus. These are the kinds of everyday experiences in which a mathematical sophisticate’s understanding of continuity is rooted. Thus we see that to understand the rich concept of *continuity* is to have built up this kind of imagery, and the epsilons and deltas are merely tools, used briefly and discarded quickly, which help one to realize that goal.

In sum, notions taught in schools and colleges are internalized by students not formally, but by means of naïve analogies, which is to say, by means of analogies with familiar categories. These categories can come from any domain, and they tend to come from domains that are not covered in school. Thus the presumption that a given area of knowledge is self-contained is erroneous, because students will inevitably connect every new notion in that area with experiences they've had in other areas of life. Moreover, a naïve analogy, once it has taken root in a beginner's mind, will not go away as time passes; it will stick in the mind as school goes on, year after year, because everyone has a deep need for simple, basic intuitions. Indeed, some naïve analogies are so persistent that one might begin to doubt whether schooling can have any effect at all in certain areas. The goal of transmitting to average pupils subtle ideas that humanity as a whole took thousands of years to discover and absorb is admirable, but it is not something that just happens by itself.

We will see that certain notions that are usually thought of as extremely simple and that are taught in elementary school are, unfortunately, understood through naïve analogies having quite limited realms of validity, and these often-misleading analogies remain entrenched through middle school, high school, and college. Their subliminal effect lives on, long after education should theoretically have thoroughly drummed them out. Thus naïve analogies, thanks to their great robustness, lie behind the thought processes of well-educated adults, even experts; this casts doubt on the idea that education gets rid of them. The stereotypical vision, by contrast, is that people who have deeply absorbed scientific ideas swim in a soup of formally defined notions that obey logical laws; analogies play no part in this vision. That, however, is just wishful thinking. To be sure, scientific education does convey ideas that go beyond everyday experience, and in particular they tend to be more abstract, but scientific ideas are not learned any differently from other ideas — they, too, are rooted in naïve analogies, and this is a universal fact, whether one is talking of beginners, intermediates, or experts.

Where Novelty and Familiarity Walk Hand in Hand

There is no question that the development of computers and related technologies has given rise to a major revolution on our planet. Those who remember the day when there were no computers in homes or businesses, and when there was no World-Wide Web to which to connect such devices, have children who are filled with wonderment at the idea that people could get along at all in such a primitive world. It thus makes sense to suggest that computers were *the* great innovation of the twentieth century. It also might seem that when something so new and so completely unprecedented takes over so widely, it would involve such a radical break with prior concepts that completely new categories would have to be introduced, categories not grounded in any kind of analogy at all.

If computer technologies are so hugely innovative, don't they have to do much more than merely use various analogies to clothe familiar old notions in fancy new wrappings? Actually, the key question is the exact opposite — namely: how could